Quantum Spectrodynamics v2.x

Errata of the v2.0 Release Set and Revisions to the v1.0 Errata

Part 1 v6, Supplement S8 v2 and Supplement S9 v3;
the v1.0 errata, revision 1

Revision 2, release v2.0.1

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Abstract

This file corrects the three documents released as v2.0 on 29 August 2026: Part 1 v6 (code selection and spacetime signature), Supplement S8 v2 (crossing errors and closed timelike curves) and Supplement S9 v3 (the observer). It replaces Revision 1 of 1 October 2026, a draft that was never published and is kept as the older version; nothing of it is carried over unchecked. The programme is being rebuilt from its axiom, every rigorous claim with a machine check, and the rebuild changes the status of the central results: the code $[\![5, 1, 3]\!]_2$ follows from the axiom under two named premises, not from the axiom alone, and the spacetime algebra $\mathrm{Cl}(1, 3)$ is carried by the code without being selected by it. Most of the mathematics stands, some of it strengthened. Four statements are contradicted outright, the triple-equivalence proof of $k = 1$ is withdrawn, the observer route to $q = 2$ is withdrawn with the apparent code it needs, which does not exist, and Supplement S8's pseudoscalar formula is wrong in general. Each row quotes the released text and states the correction with its reason. Part B lists the revisions owed to the published v1.0 errata: the rows whose status the same rulings change, the pointers and quotations that the released v1.0 sources contradict, and the corrections that wait for the rebuild of the later parts.

1 How to read this file

Status. Nothing here modifies a published text. The v2.0 set stays as released; this file is its reading layer until rebuilt papers supersede it (§6). Where this file and a released text disagree, this file wins.

Citation rule. A claim of the v2.0 set listed in Part A is cited from its corrected reading; the released text is cited for priority. A claim not listed is not thereby confirmed: what the rebuild has re-checked is listed in §4, and the rest is unverified until re-checked.

The record's DOI. The four documents of release v2.0 (Part 1 v6, S8 v2, S9 v3 and the v1.0 errata) cite the record as "Quantum spectrodynamics. Zenodo, 2025. DOI:10.5281/zenodo.19162378", and its supplementary material under the same DOI. That DOI belongs to a separate paper. The record, first published in 2026, is 10.5281/zenodo.18832272 (all versions); release v2.0 is 10.5281/zenodo.22161525.

Classes. The classes of the v1.0 errata, with one added:

[SUPERSEDED]	withdrawn as stated; never restate it
[RETAGGED]	the claim stands, with a weaker tag
[REFORMULATED]	right result, wrong reason or retired language
[DRIFT]	two passages of the set disagree
[CONFIRMED]	re-checked by the rebuild; stands as printed
[UPGRADED]	strengthened by the rebuild

Tags. [R] rigorous within QSD, each with a machine check; [C+] a conjecture whose premises are named; [T] established thermodynamics (the Second Law and Landauer's principle); [O] open.

References to the rebuild. A bracket such as [rebuild: SEL-4, R-017] names the claims of the rebuilt knowledge base and the rulings behind a row; every [R] claim there has a machine check. Each row states its reason without them; the knowledge base is not yet public and is to be released with the rebuilt papers. Quotations are verbatim, with the released documents' own macros written out; [...] marks a cut.

2 The axiom and the admitted premises

The rows below correct the v2.0 set against the axiom as it now reads and against three premises, each named where it is used.

L0, the QSD axiom. The universe is a quantum error-correcting code $[\![n, k, d]\!]_q$: it corrects at least one arbitrary error on any of its qudits, so $d \geq 3$; and to correct is to keep information, so it encodes at least one logical qudit, $k \geq 1$. All four parameters are unknown at the outset.

Admitted beyond L0: established thermodynamics [T], and these premises [C+]:

- *the crossing premise*: in a code with $k \geq 2$ the crossing operators act on the logical register as channels, and the single-qubit logical operators do not;
- *the least-dissipation premise*: the realized code has the least irreducible dissipation per correction cycle at the protection L0 requires, measured by the entropy its recovery must pass on when the noise is unknown;
- *the imaginary-unit premise*: the imaginary unit of quantum mechanics is one concept realized three ways — the complex structure of the real evolution, the spatial bivector $\gamma_2\gamma_1$ acting on spinors from the right, and the pseudoscalar $\gamma_0\gamma_1\gamma_2\gamma_3$ — taken as three routes that must agree.

Figure 1 shows where the premises enter. The v2.0 set states the whole chain as [R] "given L0"; with the premises named, its links carry the tags of the figure.

Notation note (no erratum). Part 1 prints the stabiliser generators as $S_1 = XZZXI$, $S_2 = IXZZX$, $S_3 = XIXZZ$, $S_4 = ZXIXZ$. The cyclic orbit, labelled by the position of the identity letter, is $Q_0 = IXZZX$, ..., $Q_4 = XZZXI$, so that $S_1, \ldots, S_4 = Q_4, Q_0, Q_1, Q_2$; Part 2 reads the same group in time order. The rows below use Part 1's order.

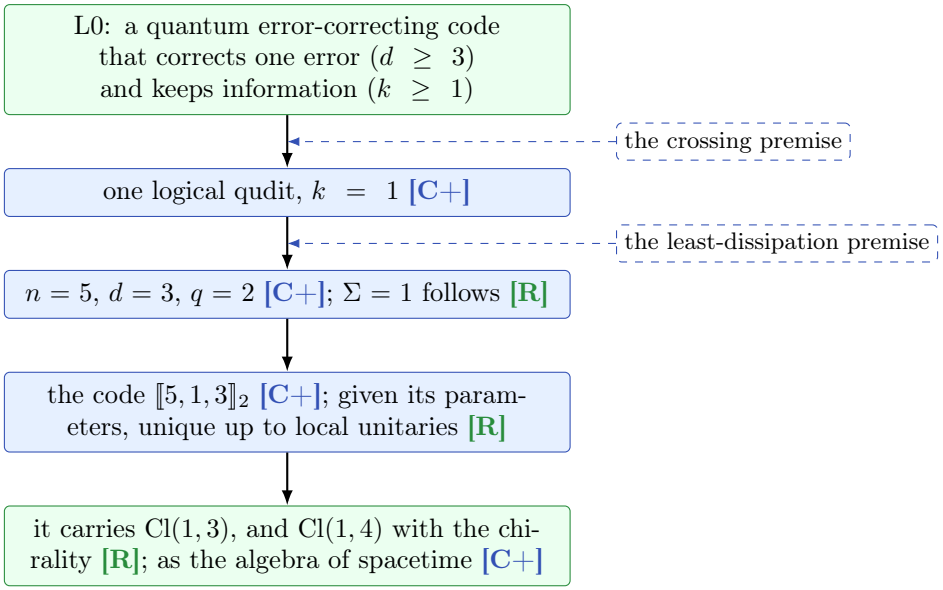


Figure 1: The selection chain as corrected. Boxes with a green frame are [R], blue ones [C+]; dashed boxes are the premises where they enter. The v2.0 set tags every link [R] and names neither premise.

3 Part A — the v2.0 set

3.1 Part 1 v6, Code Selection and Spacetime Signature [1]

item	as published	correction
P1-1	[REFORMULATED] §1, the axiom: "*The universe is a quantum error-correcting code* $[\![n, k, d]\!]_q$." §3.1: "For the code to maintain a stable logical state under continuous environmental perturbation, it must identify and correct errors. This requires $d \geq 3$." Constraint I is tagged [R].	$d \geq 3$ **is part of the axiom, not a result.** L0 reads as in §2: the code corrects at least one arbitrary error on any qudit, so $d \geq 3$, and keeps information, so $k \geq 1$. The "stable logical state" of §3.1 is this clause of the axiom, not a further requirement, and Constraint I is its reading. Beyond L0 only established thermodynamics and the premises of §2 are admitted. [rebuild: R-014, R-016, R-028]
P1-2	[REFORMULATED] §3.2: "With $k = 0$ the code encodes nothing; we therefore have $k \geq 1$."	**The conclusion stands; its reason is the axiom.** Coding theory does not exclude $k = 0$: a code space of dimension one meets the error-correction conditions for every error, so one qubit reset to $\|0\rangle$ "corrects" everything, and its recovery passes only $\ln 2$ per cycle, less than the five-qubit code's $4 \ln 2$. Neither the bounds nor least dissipation exclude it; $k \geq 1$ is part of L0, since to correct is to keep information. The same holds for Supplement S8 §6 (S8-11). [rebuild: R-028, SEL-8]

item	as published	correction
P1-3	**[REFORMULATED]** §3.2: "Each crossing error induces a quantum channel M_s on the joint logical state, and the channel admits the Deutsch fixed-point equation [...] The unique solution is the maximally mixed state $\rho^* = I/2^k$, from which no information can be extracted."	**Unique for the crossing sector, not for one crossing operator.** The channel of one operator such as $\bar{X}_i\bar{X}_j$ leaves an eight-dimensional set of fixed operators on two logical qubits. The nine logical operators that act on both qubits of a pair, the pair's crossing sector, together generate the whole logical operator algebra; a channel that applies all of them, with any positive weights, has exactly one fixed state, $\mathbb{I}/2^k$, for every $k \geq 2$, for registers of qubits or of qudits, in any code: neither n nor d enters (Fig. 2). That this state is the fixed point of a closed-timelike-curve channel in Deutsch's model [5] is Supplement S8's reading; the corrected argument for $k = 1$ does not use it (P1-4). [rebuild: CS-1, CS-2, CS-6]
P1-4	**[RETAGGED]** §3.2, box: "Constraint II **[R]**. [...] $k = 1$. Causal consistency uniquely selects one logical qudit. The construction above maps k logical qudits to k timelike directions;"	**[C+], on the crossing premise.** The step needs a premise Part 1 does not state: in a code with $k \geq 2$ the crossing operators act on the logical register as channels, and the single-qubit logical operators do not. The second half is needed: applied as a channel, the single-qubit logical operators also leave only $\mathbb{I}/2$ fixed, and without it the same argument would exclude $k = 1$ as well. Under the premise, a register driven to $\mathbb{I}/2^k$ keeps no earlier state to restore, against the axiom's requirement to correct; hence $k = 1$, for every code $[\![n, k, d]\!]_q$ (P1-3). The [C+] premise of the same section on a global complex structure is not needed for this step; "maps k logical qudits to k timelike directions" is a definition (P1-5). [rebuild: R-019, CS-4, CS-5]
P1-5	**[REFORMULATED]** §3.2: "The pathologies of $k \geq 2$ are not three accidents but three faces of one structural fact: [...] Only $k = 1$ avoids all three."	**Not the proof of $k = 1$; its faces stand separately.** The triple equivalence states that logical content and timelike direction are one thing: a code with k logical qudits carries k timelike directions. Of its faces, (i) a crossing error drives the register to $\mathbb{I}/2^k$, proved for every $k \geq 2$ (P1-3); (ii) one timelike generator $i\bar{Z}$ per logical qudit is a definition, since the operator algebra fixes neither the number of timelike generators nor the signature (Part 1 §6 says so itself), so this face is open; (iii) among all signatures of the code's five Clifford generators (P1-14), a central imaginary unit with no plane spanned by two timelike directions occurs only with one timelike generator, proved with t, the number of timelike generators, in place of k. Supplement S8's proof is withdrawn (S8-7). [rebuild: R-030, CS-1, CL-7, CL-O1]

item	as published	correction
P1-6	**[RETAGGED]** §3.4: "Taking the minimum at equality, $n = 5$, the code $[\![5, 1, 3]\!]_q$ is a valid stabiliser code for every prime q." §3.5: "Both $d = 3$ and $q = 2$ are derived consequences of double extremality given $k = 1$, not independent assumptions."	**[C+], on the least-dissipation premise.** Nothing in Part 1 justifies taking either bound at equality; double extremality is an unstated premise. What replaces it is the least-dissipation premise (§2). Any recovery of a code of dimension K on n qudits passes at least $\ln(q^n/K)$ to its environment when the noise is unknown [6], which is $(n-k)\ln q$ for $K = q^k$; with the quantum Singleton bound $n - k \geq 2(d-1)$, which holds for every quantum code [8, 9], the least cost at $k = 1$, $d \geq 3$ is $4\ln 2$, reached only at $n = 5$, $d = 3$, $q = 2$. Singleton equality is then a result, and Hamming equality ($\Sigma = 1$) a consequence and a check, not a selecting step. [rebuild: R-017, R-020, R-024, SEL-2, SEL-3, SEL-7]
P1-7	**[REFORMULATED]** §2: "Two fundamental inequalities constrain the parameters of any quantum stabiliser code $[\![n, k, d]\!]_q$." §2: "The *quantum Hamming bound* [...] for a code correcting up to t errors ($d = 2t + 1$), requires"	**The selection holds among all codes, over any alphabet.** The derivation is stated for stabiliser codes over a prime alphabet, while the axiom and the Selection Theorem speak of every quantum code; and the quantum Hamming bound is used without its condition (it is proved for non-degenerate codes). Neither restriction is needed: the floor $\ln(q^n/K)$ holds for any code and any recovery, and the Singleton bound for every quantum code, so the least-dissipation selection (P1-6) runs over all codes and alphabets. The Hamming bound enters only as a check. [rebuild: R-018, R-024, SEL-4, SEL-7]
P1-8	**[REFORMULATED]** §3.4: "Because the Singleton bound [...] depends only on $\tau = 2$ (through the syndrome overhead $n - k = 2(d-1)$) and the service level d, it contains no q-dependence whatsoever. With $k = 1$ from §3.2 and $d = 3$ from §3.1,"	**A reading, and a step out of order.** The quantum Singleton bound holds for every quantum code with its factor 2, whatever that factor is read as; reading it as $\tau = 2$, the two independent error axes of every prime (which hold, §4), is an interpretation. The step uses $d = 3$ before §3.5 fixes it, since §3.1 gives $d \geq 3$; under least dissipation $n = 5$ and $d = 3$ follow together (P1-6). [rebuild: ERR-2, SEL-2]
P1-9	**[RETAGGED]** §3.4, box: "Universal native code **[R]**. [...] For every prime q, the unique minimal CTC-free single-error-correcting stabiliser code is $[\![5, 1, 3]\!]_q$."	**[R] for the code at every prime; "unique" only at $q = 2$.** Part 1 gives no construction for $q > 2$. The rebuild proves one for every prime: the cyclic shifts of $(X, Z, Z^{-1}, X^{-1}, I)$ commute, and an error on any two qudits has a syndrome given by a 4×4 integer matrix of determinant ± 1, so they give a code $[\![5, 1, 3]\!]_q$ with distinct syndromes for all correctable errors (§4). Its length is minimal by the quantum Singleton bound, $n \geq 5$ at $k = 1$, $d = 3$, which holds for every code (P1-6). Uniqueness is proved only for $q = 2$ (P1-12), and "CTC-free" imports $k = 1$ (P1-4). [rebuild: ERR-4, SEL-1, SEL-2, SEL-5]

item	as published	correction
P1-10	**[REFORMULATED]** §3.5: "For $d \geq 6$, the Hamming gap widens monotonically with d at every prime q." §3.5, box: "For all $d \geq 4$, no prime-power solution exists."	**Checked to $d = 10$, and no longer needed.** Part 1 computes $d = 4$ and $d = 5$ (the latter only at $q = 2$) and asserts the rest. A parallel check finds no integer root $q \geq 2$ of the Hamming equality at $n = 1 + 2(d-1)$, $k = 1$ for $d = 4$ to 10; beyond that the claim is unproved. It is not needed: $d = 3$ follows from least dissipation (P1-6). [rebuild: ERR-5]
P1-11	**[RETAGGED]** abstract: "We derive the complete identity of the universe's quantum error-correcting code from a single axiom." §4, box: "The QECC Selection Theorem **[R]**. [...] Given L0, the code is $[\![5, 1, 3]\!]_2$. No other QECC is consistent with L0." §10: "*No parameter is assumed.*"	**[C+], on two named premises; two sentences withdrawn.** $[\![5, 1, 3]\!]_2$ follows from L0 under the crossing premise ($k = 1$, P1-4) and the least-dissipation premise ($n = 5$, $d = 3$, $q = 2$, P1-6), among all quantum codes over any alphabet (P1-7). "No other QECC is consistent with L0" is **withdrawn**: other codes with $k = 1$ and $d \geq 3$ satisfy L0 and pass more entropy per cycle (a code with $n = 6$, $k = 1$, $d = 3$ passes $5 \ln 2$). "No parameter is assumed" is **withdrawn**: the two premises carry k and n, d, q. The same reading applies to the title and abstract ("uniquely determine"), §1 ("We derive every one of them"), §3 ("Six sequential constraints"), §4 ("from structural constraints alone"), the caption of the selection funnel ("neither is assumed") and the summary table of §10. [rebuild: R-017, R-019, R-024, SEL-4]
P1-12	**[UPGRADED]** §4: "among all $[\![5, 1, 3]\!]_2$ stabiliser codes, the cyclic-shift code is unique up to local Clifford equivalence."	**Checked, and strengthened.** An exhaustive search confirms it: of the 782 595 groups of four commuting generators on five qubits, exactly 2 592 have $d = 3$, and all are the cyclic code under local Cliffords. Among all quantum codes, stabiliser or not, a code with the parameters of the five-qubit code is unique up to local unitaries [10] (used, not re-proved). [rebuild: SEL-5, SEL-6, R-023]
P1-13	**[RETAGGED]** §5.3, box: "The Second Law of Thermodynamics and Landauer's principle (both empirically established) require that any physical code with $\Sigma < 1$ dissipates irreducible entropy per correction cycle." §5.3, box: "This is **[T]**: not an axiom of QSD, not a conjecture, but a mandatory consistency condition with known physics."	**[C+], not [T].** The Second Law and Landauer's principle fix what a correction costs; they do not require the realized code to be the cheapest. That step is the least-dissipation premise. A code with $\Sigma < 1$ dissipates more per cycle but violates no established law. Under the premise $q = 2$ follows (P1-6), and $\Sigma(2) = 1$ is a consequence; $\Sigma(q) < 1$ for $q \geq 3$, the ghost slots, stands as computed (§4). The comparison of §5 runs over $[\![5, 1, 3]\!]_q$ at fixed $n = 5$; the premise compares all codes. The same reading applies to the abstract, §8 (P1-17), the caption of §9 ("Only $q = 2$ is thermodynamically stable") and the summary row of §10. [rebuild: R-016, R-017, SEL-3, SEL-4]

item	as published	correction
P1-14	**[REFORMULATED]** §6, box: "The code's operator algebra carries $\mathrm{Cl}(1,3)$ exactly: one timelike direction from the logical qubit, three spacelike directions realized in the error sector, with the explicit generator set exhibited in Part II."	**The set is exhibited here; the code carries $\mathrm{Cl}(1,4)$.** Part 2 does not exhibit the generator set either. It is $\gamma_0 = i\bar{Z}$, $\gamma_1 = \bar{X}\mathrm{D}_1$, $\gamma_2 = \bar{X}\mathrm{S}_1\mathrm{D}_2$, $\gamma_3 = \bar{X}\mathrm{S}_1\mathrm{S}_2\mathrm{D}_3$, with the destabilisers $\mathrm{D}_1 = IIIYY$, $\mathrm{D}_2 = IXXII$, $\mathrm{D}_3 = IZZXX$; its Gram matrix is $\mathrm{diag}(-1,+1,+1,+1)$ exactly **[R]**. With $\mathrm{D}_4 = ZIXIY$ the ladder continues to a fifth generator, $\gamma_4 = \bar{X}\mathrm{S}_1\mathrm{S}_2\mathrm{S}_3\mathrm{D}_4$: the code carries $\mathrm{Cl}(1,4)$. The fifth is not a further direction: the product of all five, $P_5 = -i\,IYYYX$, commutes with all of them and squares to -1, and $\gamma_4 = -P_5\,\gamma_0\gamma_1\gamma_2\gamma_3$, the chirality operator of the Dirac algebra with P_5 in the role of i; $\mathrm{Cl}(1,4) = \mathrm{Cl}(1,3) \oplus P_5\,\mathrm{Cl}(1,3)$ **[R]**. As §6 itself says, the algebra fixes neither the number of spacelike directions nor the signature; that the universe's code carries $\mathrm{Cl}(1,3)$ as the algebra of spacetime is **[C+]**. The title ("Uniquely Determine … $\mathrm{Cl}(1,3)$") reads accordingly. [rebuild: CL-1, CL-2, CL-4, CL-5, CL-6]
P1-15	**[SUPERSEDED]** §6, figure caption: "The algebra family $\mathrm{Cl}(k,4-k)$ in context. The active phase $\mathrm{Cl}(1,3)$ is the unique level with $\mathcal{I}^2 = -1$."	**The family is not derived.** No section of Part 1 derives $\mathrm{Cl}(k,4-k)$, and by §6 the signature is not an operator-algebra fact. For four anticommuting generators, t of them squaring to -1, the pseudoscalar squares to $(-1)^t$, so $\mathrm{Cl}(3,1)$ has $\mathcal{I}^2 = -1$ as well, and "unique level" holds only among the three algebras drawn. The flanking phases are left to the later parts. [rebuild: CL-4, CL-7]
P1-16	**[RETAGGED]** §7, box: "Complex structure **[R]**. […] Given the code-supplied $\mathrm{Cl}(1,3)$, the imaginary unit of quantum mechanics is the spin-plane bivector $\gamma_2\gamma_1$ — a theorem of the geometry, not a postulate."	**A premise, not a theorem: [C+].** That $\gamma_2\gamma_1$ is quantum mechanics' i is the imaginary-unit premise (§2). What is proved **[R]**: in four dimensions the Dirac equation can be written and solved with real numbers only, so an imaginary unit is extra structure; every real rotation carries one, the complex structure J of its evolution; J equals right multiplication by $\gamma_2\gamma_1$ on positive-energy solutions; the pseudoscalar agrees with both only for massless solutions of one chirality. In the code, $\gamma_2\gamma_1 = -i\,XYYZY$ acts in the error sector, not on the logical qubit. "A theorem of the geometry, not a postulate" is **withdrawn**, here and in the abstract ("follows geometrically, not by postulate"); the summary row "Complex structure: $\mathcal{I}^2 = -1$" of §10 holds for the sign, not for the role. [rebuild: R-022, CX-1, CX-3, CX-4, CX-5, CX-6, CX-7]

item	as published	correction
P1-17	**[DRIFT]** §8: "The $q = 2$ observer is the unique thermodynamically stable ground state **[T]** ($\Sigma = 1$)." §8, box: "The $q = 2$ observer is the unique thermodynamically stable observer ($\Sigma = 1$)." §8: "The QECC structure is not postulated — it follows from stable memory ($d \geq 3$) and definite outcomes ($k = 1$ per measurement)."	**One claim, two tags; [C+].** The same claim is tagged **[T]** in the definition and **[R]** in the box (headed "Observer **[R]**"). It rests on §5.3 and, read with P1-13, on the least-dissipation premise: **[C+]**. That the observer's register corrects errors is not postulated and does not follow from stable memory: an observer is a subsystem of the universe's code that observes its environment from within, so the axiom applies to it, $d \geq 3$ and $k \geq 1$ included (S9-1). That its register is $[\![5, 1, 3]\!]_2$ follows only by applying the crossing and least-dissipation premises to it: **[C+]**. [rebuild: OBS-5, R-017, R-033]
P1-18	**[SUPERSEDED]** §10: "The number of spatial dimensions — three — is a theorem, not an input."	**Struck.** It contradicts §6, which says that neither the number of spacelike directions nor the signature is an operator-algebra fact. Three is the count of the exhibited generator set; the code also carries a fifth generator, the chirality (P1-14). Where the count of spacetime directions comes from is the question of the node (Part 2). Revision 1 struck the sentence as well. [rebuild: CL-4, CL-6]
P1-19	**[SUPERSEDED]** §3.6: "The same algebraic mechanism that selected $k = 1$ in §3.2 eliminates every $q \geq 3$ universe for any $q = 2$ observer." §4: "Each is sufficient on its own."	**Withdrawn.** §3.6 names its premise (binary encoding of qudits) and tags the route **[C+]**, but the route fails twice. The apparent code it leads to does not exist: no qubit code of dimension 2^{n-4} with $d = 3$ exists for $n = 6$ to 20, stabiliser or not, so neither $[\![10, 6, 3]\!]_2$ ($q = 3$) nor $[\![15, 11, 3]\!]_2$ ($q = 5, 7$) does. And an observer is a subsystem of the universe's code, made of its qudits (S9-1), so there is no qubit observer of a universe with $q \geq 3$. Its repetitions in the abstract, §1 and §4 are withdrawn with it. The corrected selection never used the route: $q = 2$ follows from least dissipation (P1-6). [rebuild: ERR-6, R-033, R-036, SEL-4]

item	as published	correction

3.2 Supplement S8 v2, The Crossing Error–CTC Correspondence [2]

item	as published	correction
S8-1	**[SUPERSEDED]** §1: "every CPTP channel is a contraction in the trace norm, [. . .] with strict contraction toward ρ^* for any non-unital component. Iterating $M_{\mathbf{s}}$ on any input drives the state toward the unique maximally mixed fixed point."	**The stated reason is false twice over.** A unitary channel preserves the trace distance exactly, and a non-unital channel need not fix $\mathbb{I}/2^k$ at all (amplitude damping fixes $\|0\rangle\langle 0\|$). What is true: $\mathbb{I}/2^k$ is fixed by every unital channel; the fixed operators of a unital channel are those that commute with all its Kraus operators [7]; so the fixed state is unique exactly when the Kraus operators generate the whole operator algebra. Uniqueness is a property of which operators act, not of iteration (S8-2). The box of §1 ("derived above without further premises") reads with this. [rebuild: CS-1, CS-2]
S8-2	**[SUPERSEDED]** §2: "For each single-qubit factor, the only state invariant under conjugation by X is the maximally mixed $I/2$."	**False for one operator.** The operators commuting with X are spanned by I and X, and conjugation by $\bar{X}_i\bar{X}_j$ leaves an eight-dimensional set of fixed operators on two logical qubits; the channel of §2 is also written only on product states $\rho_i \otimes \rho_j$. What is true **[R]**: the crossing sector of the pair, all nine $\bar{L}_i\bar{L}_j$ with $\bar{L} \in \{\bar{X}, \bar{Y}, \bar{Z}\}$, generates the whole two-qubit logical algebra, and its joint fixed state is unique, $\mathbb{I}/4$; adding its operators one at a time, the fixed-point dimension falls $8 \to 4 \to 2 \to 2 \to 1$ (Fig. 2). For $k \geq 2$ the sector's fixed state $\mathbb{I}/2^k$ is unique, for qubits and qudits. Every "unique fixed point" of S8 (abstract, §§1–2, 4.2, 4.3 and 5) is to be read for the sector, not for one operator. [rebuild: CS-1, CS-2, CS-6]
S8-3	**[RETAGGED]** §2: "The unitary action of C_{ij} induces a channel on the joint logical state $\rho_L = \rho_i \otimes \rho_j$:" §7: "All results are **[R]**— derived from the stabiliser formalism and the Deutsch mechanism without additional premises beyond L0."	**The $k = 1$ results are [C+], on the crossing premise.** The step from "a crossing operator exists in the code" to "it acts on the logical register as a channel" is a premise, already used in the abstract and §1 and never named: the crossing premise (§2), with its second half, that the single-qubit logical operators do not act so (P1-4). The statements about fixed points stay **[R]** in the sector reading (S8-2); the results that conclude $k = 1$, and the status statements of the context box, the summary table and §7, are **[C+]**. [rebuild: R-019, CS-4, CS-5]
S8-4	**[REFORMULATED]** §2, box: "The formal identity is exact, not an analogy: physical register ≡ CTC region, logical qudits ≡ system traversing the loop."	**Exact is the fixed state; the identification is a reading.** What is exact: the crossing sector's unique fixed state is the state that Deutsch's consistency condition assigns to a system on a closed timelike curve, $\mathbb{I}/2^k$ [5]. That the crossing channel *is* such a curve is a reading, not used by the corrected argument for $k = 1$ (S8-3). The caption of Fig. 1 assigns the roles the other way round (the logical pair as the chronology-respecting system, the register as the curve). [rebuild: CS-1, R-019]

item	as published	correction
S8-5	**[SUPERSEDED]** §3: "Cl(k, $n-2k$): k timelike generators (one from each logical Z via $\gamma_0 = i\bar{Z}$, which satisfies $\gamma_0^2 = -I$), and $n-2k$ spacelike generators (from the residual stabiliser content after the timelike directions are subtracted)."	**The mechanism fails, and the count is withdrawn.** The stabiliser generators commute with one another and with the logical operators, so no residual stabiliser content can supply anticommuting generators. For one logical qubit the destabiliser-dressed logical operators give Cl$(1, n-1)$: Cl$(1,4)$ for the five-qubit code, whose fifth generator is the chirality, not a fifth direction (P1-14). The count $n-2k$ is withdrawn as a general rule, and with it Revision 1's statement that the count is unaffected. One timelike generator per logical qudit is a definition: the algebra fixes neither the number of timelike generators nor the signature (P1-5). The subtitle ("Why Logical Qubits Count Timelike Directions") reads accordingly. [rebuild: CL-2, CL-4, CL-6, CL-O1, DR-014]
S8-6	**[SUPERSEDED]** §3: "The pseudoscalar of Cl(p,q), where p counts timelike and q counts spacelike generators, satisfies" $\mathcal{I}^2 = (-1)^{p(p-1)/2}(-1)^q$ (eq. 10); §3, box: "The pseudoscalar of the syndrome Clifford algebra satisfies $\mathcal{I}^2 = -1$ if and only if $k = 1$"	**The formula is wrong in general.** Reversing a product of $N = p+q$ anticommuting generators costs $(-1)^{N(N-1)/2}$, so $\mathcal{I}^2 = (-1)^{N(N-1)/2}(-1)^p$. The two agree for Cl$(1,3)$ but not in general: for S8's own $k=2$, $n=5$ assignment, Cl$(2,1)$, $\mathcal{I}^2 = -1$, not $+1$; for Cl$(2,2)$ it is $+1$, for Cl$(1,4)$ it is -1. So the "$+1$ at $k=2$" of §3, the box "Clifford pathology", condition (C3) of §5 and the step (C3)⇒(C1) do not follow even from S8's own assignment. For the five generators of the five-qubit code with t of them timelike, the product squares to $(-1)^t$ **[R]**. [rebuild: CL-7]
S8-7	**[SUPERSEDED]** §5: "For any stabiliser code $[\![n,k,d]\!]_q$ (with $n=5$, $d=3$ as fixed by P1 §3.4–3.5), the following three conditions are mutually equivalent:" §5, box: "All three are absent if and only if $k=1$. The code $[\![5,1,3]\!]_2$ is therefore the unique CTC-free, computationally consistent, QM-supporting stabiliser code selected by Part I."	**Withdrawn as a theorem and as the proof of $k=1$.** (i) $n=5$ and $d=3$ come from Part 1 §§3.4–3.5, which take $k=1$ as given: the proof is circular. (ii) Within its own hypothesis the quantum Singleton bound $n-k \geq 2(d-1)$ gives $k \leq 1$, so (C1) never holds and the equivalences are vacuous; the $k=2$ and $k=6$ examples lie outside it. (iii) (C3) rests on the pseudoscalar formula (S8-6) and on one timelike generator per logical qudit, a definition (S8-5). (iv) The theorem does not select $q=2$: the five-qutrit code also has $n=5$, $d=3$, $k=1$. The triple equivalence is read as in P1-5, logical content and timelike direction as one thing, with its faces proved or open separately; $k=1$ rests on the crossing premise (S8-3). [rebuild: R-030, R-019, CS-1, CL-7, CL-O1]
S8-8	**[RETAGGED]** §4.1: "The five-qubit code is the unique survivor of the QECC Selection Theorem of P1 §4." §4.1, box: "$[\![5,1,3]\!]_2$ has zero crossing pairs, $\mathcal{I}^2 = -1$, a unique γ_0-split, and a syndrome table in perfect bijection with the correctable errors."	**The parts hold; the selection is [C+].** No crossing pairs ($k=1$), a Cl$(1,3)$ generator set whose pseudoscalar squares to -1, and sixteen syndromes in bijection with the identity and the fifteen single-qubit errors: **[R]**. The signature is a choice (a factor i flips any generator's square), $i = \gamma_2\gamma_1$ is one route of the imaginary-unit premise (P1-16), and the selection of the code is **[C+]** (P1-11). [rebuild: CL-1, CL-4, SEL-3, SEL-4]

item	as published	correction
S8-9	**[DRIFT]** §4.2: "Take any $[\![n, 2, d]\!]_2$ stabiliser code with $d \geq 3$ for concreteness; the standard $[\![4, 2, 2]\!]$ Bell-pair code is the minimal example."	$[\![4, 2, 2]\!]$ **has $d = 2$.** It is not an example of the case announced. The crossing sector does not depend on d: on $[\![4, 2, 2]\!]$, and on a code $[\![8, 3, 3]\!]$ with $d = 3$, every crossing operator is logical, and the encoded state decays to the maximally mixed state while the syndrome reads trivial in every round. The box of §4.2 is **[C+]** on the crossing premise (S8-3). [rebuild: CS-3]
S8-10	**[SUPERSEDED]** §3: "the relevant pseudoscalar is that of $\mathrm{Cl}(6, 4)$" §4.3: "The apparent algebra is $\mathrm{Cl}(6, n_{\mathrm{app}} - 12)$ with $n_{\mathrm{app}} - 12 = -2$"	**Two algebras for a code that does not exist.** §3 and §4.3 assign different algebras to the same apparent code, and neither follows from an operator construction (S8-5). The code itself, $[\![10, 6, 3]\!]_2$, does not exist, and the observer route it serves is withdrawn (P1-19): the box "$k = 6$: 15 CTCs" and its summary row are **withdrawn** with it. [rebuild: ERR-6, R-036]
S8-11	**[REFORMULATED]** §6: "$k = 0$ is excluded by L0, which posits the universe to be a code with observable content — the universe contains something rather than nothing."	**Stands, with the axiom as stated here.** L0 includes $k \geq 1$, since to correct is to keep information (P1-2). The theorem's "absent if and only if $k = 1$" needs this, for all three conditions are also absent at $k = 0$. [rebuild: R-028, SEL-8]

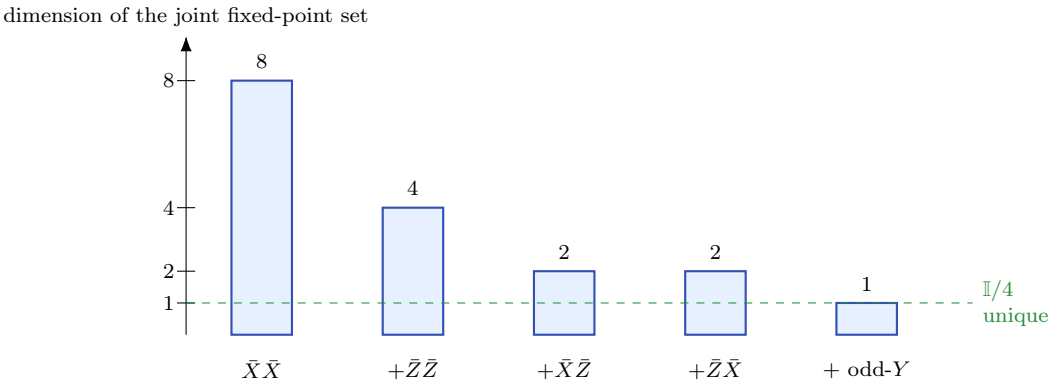


Figure 2: S8-2 and P1-3: the dimension of the set of operators fixed by the crossing sector of two logical qubits, as its operators are added in this order: $8 \to 4 \to 2 \to 2 \to 1$. After four, $\bar{Y}\bar{Y}$ still commutes with all of them; one operator with an odd number of Y finishes it. Uniqueness is a property of the sector's operator content, not of iterating one channel.

3.3 Supplement S9 v3, The Observer [3]

item	as published	correction
S9-1	**[DRIFT]** §1: "The QECC structure is not assumed. It follows from what observation *is*: the production and storage of classical records under environmental perturbation." §2, box: "*Premise: records are quantum registers under single-qudit Pauli noise.* Then an observer's memory register requires $d \geq 3$ for stable record-keeping — the same error-correction constraint derived for the universe's code in Part I, §3.1"	**It follows from the axiom, not from observation.** §1 and the abstract say that the structure "follows", with "no postulate added"; §2, the definition box and the summary table tag the same constraints **[C+]** with named premises. The tags stand for $k = 1$ and for the code, not for $d \geq 3$. For the universe's code $d \geq 3$ is part of the axiom, not derived in Part 1 §3.1 (P1-1). For an observer it is the axiom too, and S9's noise premise is not needed: an observer is a subsystem of the universe's code that observes its environment from within, so the axiom applies to it, and its memory has at least five qubits, a node or a set of nodes. A single qudit of a node holds no protected record: no one or two qubits of the five-qubit code carry anything of it. Its $k = 1$ and the code $[\![5, 1, 3]\!]_2$ stay **[C+]** on the crossing and least-dissipation premises. [rebuild: R-014, R-033, SEL-1, OBS-3, OBS-5]
S9-2	**[RETAGGED]** §3, box: "The underlying triple equivalence ($\binom{k}{2} \geq 1 \Leftrightarrow$ Deutsch channel $\Leftrightarrow \mathcal{I}^2 = +1$) is algebra and remains **[R]**" §3: "and $\mathcal{I}^2 = -1$ in the [...] $\mathrm{Cl}(1, n-2)$."	**Not [R].** The triple equivalence is withdrawn as a theorem (S8-7) and read as in P1-5. The observer's $k = 1$ rests on S9's own named premise, that the crossing channels are realised under the observer's correction cycles: the crossing premise applied to the observer, **[C+]** as tagged. The count $\mathrm{Cl}(1, n-2)$ is not an operator-algebra fact (S8-5); the product of the five-qubit code's five generators squares to -1 (P1-14). The summary row "S8 triple equivalence itself" (**[R]**) is withdrawn. [rebuild: R-030, CL-6, CL-7]
S9-3	**[REFORMULATED]** §3: "the simplest crossing operator is $C_{12} = \bar{X}_1 \bar{X}_2$." §3: "which has Deutsch fixed point $\rho^* = I/4$ (Supplement S8, §2)."	**The whole sector, not one operator.** Read with S8-2: $\bar{X}_1 \bar{X}_2$ alone leaves an eight-dimensional set of fixed operators; the records are driven to $\mathbb{I}/4$ by the pair's whole crossing sector. That a $k \geq 2$ observer "would equilibrate its joint record to $I/2^k$ under any internal correction cycle" (§3) holds under the premise S9 names. [rebuild: CS-1, CS-2]
S9-4	**[RETAGGED]** §4: "This is **[T]**, not an additional QSD assumption about observers and not a conjecture: a $q \geq 3$ observer would violate the Second Law in the operational sense that its accumulated irreducible entropy would diverge with operation time."	**[C+], not [T].** As in P1-13: a register with $q \geq 3$ dissipates more per cycle but violates no law; that the observer's register is the least-dissipating one is the least-dissipation premise. The same applies to the abstract, the definition box, the box of §4 and the summary row. [rebuild: R-017, SEL-3, OBS-5]
S9-5	**[RETAGGED]** §4: "because the observer is itself a $[\![5, 1, 3]\!]_q$ code block." §8: "It is a $[\![5, 1, 3]\!]_2$ node, identical in structure to every other node in the lattice,"	**Asserted, not shown: [C+] and [O].** S9 applies Part 1's argument to the observer for d and q but never for $n = 5$. That the observer's register is $[\![5, 1, 3]\!]_2$ follows only by applying the crossing and least-dissipation premises to it, **[C+]**. That it is a node of the lattice waits for the node (Part 2), not yet rebuilt: **[O]**. [rebuild: OBS-5, QSD-O2]

item	as published	correction
S9-6	**[SUPERSEDED]** §6: "the four spatial vertices form the K_4 subgraph whose automorphism group is S_4; permuting these vertices is a gauge transformation of the observer-environment split." §6, box: "The split is anchor-dependent (gauge under S_4), but all physical predictions — syndrome probabilities, post-measurement states, and correction decisions — are S_4-gauge-invariant."	**Only with local Cliffords, and up to a relabelling.** The permutations of the five qubits that preserve the code by themselves are the ten of the dihedral group D_5, of which two fix a qubit. With a local Clifford allowed on each qubit, all 120 permutations preserve the code, in three ways each, and the 24 that fix a qubit form S_4 **[R]**. Each such symmetry carries syndrome projectors onto syndrome projectors, so every prediction is unchanged *up to a relabelling of the sixteen outcomes*, not unchanged. The anchor is a construct of Part 2, not yet rebuilt. [rebuild: OBS-4]
S9-7	**[SUPERSEDED]** §7: "For the logical content of an observer $\mathcal{O}_1$ to be observed, a second observer $\mathcal{O}_2$ must perform syndrome measurements on $\mathcal{O}_1$'s physical register" §7: "with the modern consistency constraints on observers modelling observers [. . .] avoided precisely because $\mathcal{O}_1$'s record is algebraically inaccessible to $\mathcal{O}_1$"	**Syndromes cannot reveal the record.** The stabiliser generators commute with every logical operator (§7's own premise for self-blindness), and every two-qubit part of a code state of the five-qubit code is exactly $\mathbb{I}/4$. A second observer reads the record by measuring a logical operator of weight at least three on the first observer's block. Self-blindness and "self-measurement requires a second observer" (box of §7) stand with this mechanism **[R]**, as does the cut at the code boundary. That the Frauchiger–Renner constraints are thereby avoided is **withdrawn**: it is asserted without argument, and the reading it rests on is the one corrected here. [rebuild: OBS-2, OBS-3, R-034]
S9-8	**[DRIFT]** Abstract: the released abstract has no epistemic-status paragraph; the working copy's was dropped in the re-save of 26 August 2026.	**Not to be restored as it was.** The dropped paragraph tagged the S_4 gauge of the split and the triple equivalence **[R]**, and $q = 2$ **[T]**; S9-2, S9-4 and S9-6 correct all three. The statuses of S9 are those of its summary table as corrected here. Revision 1 called this a presentation matter with no content change; that is withdrawn.

4 What stands

The results the rebuild has re-checked and confirmed, each with a machine check. A result of the v2.0 set not listed here or in Part A is unverified.

result	where	class	rebuild
Every prime q has exactly two independent error axes, $\tau = 2$	Part 1 §3.3	**[CONFIRMED]**	ERR-1, ERR-2
No code $[\![n, 1, 3]\!]_2$ with $n \leq 4$: five qubits are the fewest	Part 1 §3.4	**[CONFIRMED]**	SEL-1
For every prime q the cyclic shifts of $(X, Z, Z^{-1}, X^{-1}, I)$ give a code $[\![5, 1, 3]\!]_q$, with distinct syndromes for all correctable errors	Part 1 §3.4	**[CONFIRMED]**	ERR-4

result	where	class	rebuild
Sixteen syndromes in bijection with the identity and the fifteen single-qubit errors, $\Sigma(2) = 1$; $\Sigma(q) = (5q^2 - 4)/q^4 < 1$ for $q \geq 3$, the ghost slots	Part 1 §§3.5, 5.2	**[CONFIRMED]**	SEL-3, ERR-4, ERR-5
The five-qubit code is unique: up to local Cliffords among stabiliser codes, up to local unitaries among all codes	Part 1 §4	**[UPGRADED]** (P1-12)	SEL-5, SEL-6
$\gamma_0 = i\bar{Z}$ squares to -1 and acts inside the code space; each spacelike generator carries the code space into a syndrome sector	Part 1 §6	**[CONFIRMED]**	CL-1, CL-3
Neither the number of spacelike directions nor the signature is an operator-algebra fact	Part 1 §6	**[CONFIRMED]**	CL-4
For odd q no logical operator squares to the identity, so there is no timelike generator	Part 1 §7	**[CONFIRMED]**	CX-2, ERR-3
A crossing sector has a unique fixed state, $\mathbb{I}/2^k$, for every $k \geq 2$, qubits or qudits, in any code	Supplement S8 §2	**[UPGRADED]** (S8-2)	CS-1, CS-2, CS-6
Measurement is syndrome extraction: Born weights and the Lüders update apply to the sixteen syndrome sectors, given quantum mechanics	Supplement S9 §5	**[CONFIRMED]**	OBS-1
An observer cannot read its own record with its own generators (self-blindness)	Supplement S9 §7	**[CONFIRMED]**	OBS-2
The syndrome recovery of the five-qubit code passes $4 \ln 2$ per cycle, the floor for any recovery of that code	Supplement S9 §5	**[CONFIRMED]**	SEL-7
On the code's stabiliser states every syndrome weight is 0 or 2^{-j}; continuous weights come only from the measured state	Supplement S9 §5	**[CONFIRMED]**	OBS-1

5 Part B — revisions owed to the published v1.0 errata

The v1.0 errata, revision 1 [4], corrects the v1.0 release (the main preprint and Supplements S1–S7); it was published with release v2.0 and stays as published, and the rows below are applied to it at the merge (§6). The v1.0 documents carry no epistemic tags: they call results derived, constructed or speculative in words. The tags in the v1.0 errata were assigned there, in its "epistemic retrofit" (its row E10); where a row below changes a tag, it changes that assignment, not a tag of the v1.0 texts. Revision 1 of this file audited the v1.0 errata against the released sources; its findings are taken up here as far as the rebuild reaches. B.1 holds the rows that the corrections of Part A require, B.2 the pointers and quotations checked against the released v1.0 sources, B.3 the corrections that rest on parts not yet rebuilt, listed without being asserted, and B.4 the housekeeping of the merge. In the tables, "the v1.0 errata prints" quotes the published errata's own row.

5.1 B.1 — Rows the corrections of Part A require

row	the v1.0 errata prints	correction
E2	**[RETAGGED]** §2, row E2: "$[\![5,1,3]\!]_2$ is *derived* by the seven-step selection chain **[R]** — with k=1 from Deutsch-CTC consistency, $d \geq 3$ from error completeness, and the thermodynamic selection by Landauer's principle carrying its own tag **[T]** (never "L0 alone"). P1 (v2.0) is the canonical statement."	**The verdict on the v1.0 text stands; the canonical statement is corrected.** As E2 says, the axiom of the v1.0 set carried the answer. But the code is not derived from L0 by a chain of rigorous steps. L0 now reads as in §2, with $d \geq 3$ and $k \geq 1$ part of it; $k = 1$ rests on the crossing premise, and $n = 5$, $d = 3$, $q = 2$ on the least-dissipation premise, both **[C+]** (P1-4, P1-6, P1-11); Landauer's principle fixes what a correction costs, not that the realized code is the cheapest (P1-13). The canonical statement is Part 1 v6 read through Part A. [rebuild: R-014, R-017, R-019, R-028, SEL-4]
M-2	**[RETAGGED]** §3, row M-2: "E2: L0 + seven-step selection chain; k, d, q derived **[R]** + Landauer **[T]**"	**As E2, corrected.** L0 with $d \geq 3$ and $k \geq 1$; $k = 1$ **[C+]** on the crossing premise; n, d, q **[C+]** on the least-dissipation premise; the Hamming bound a check, not a selecting step. Its pointer, §§1–2, is right but coarse: the axiom is in §1.1 and §2.1, §2.1: "The vacuum state of nature is protected by the $[[5,1,3]]$ quantum error-correcting code", and the Hamming premises in §§2.1–2.2. [rebuild: R-017, R-018, R-019, SEL-4]
S1-1	**[RETAGGED]** §4, row S1-1: "**[UPGRADED]** the premises are now theorems: k=1 **[R]** (Deutsch-CTC crossing argument), $d \geq 3$ **[R]** (error completeness), q=2 (Hamming + Landauer **[T]**)."	**The premises are not theorems.** $d \geq 3$ is part of the axiom; $k = 1$ rests on the crossing premise and $q = 2$ on the least-dissipation premise, **[C+]**. S1's uniqueness proof stands as the last step of the chain, and it is strengthened: every $[\![5,1,3]\!]_2$ stabiliser code is the five-qubit code up to local Cliffords, and among all quantum codes the code is unique up to local unitaries (P1-12). Its quotation is not in S1 ("smallest" occurs in none of the v1.0 texts); S1 states the result in §1.1: §1.1: "The unique minimal quantum code that corrects arbitrary single-qubit errors on one logical qubit is $[[5,1,3]]$." [rebuild: R-014, R-019, SEL-5, SEL-6]
§9 (S6)	**[RETAGGED]** §9: "each $[\![n,k,d]\!]_q$ defines a native theory and $[\![5,1,3]\!]_2$ is thermodynamically selected (Landauer **[T]**)."	**[C+], not [T].** As P1-13: the realized code is selected by the least-dissipation premise; Landauer's principle fixes only what each correction costs. That the landscape is code space stays **[O]**, as the v1.0 errata says. [rebuild: R-017, SEL-4]
§1	**[REFORMULATED]** §1: "Claims marked **[CONFIRMED]** may be cited from v1.0 directly." §13: "currently **757/757** checks"	**Confirmed by the earlier suite, not re-checked.** The rows marked **[CONFIRMED]** were confirmed by the earlier repository's validation suite, which the rebuild has not re-run; they may be cited from v1.0 as confirmed by that suite. Of the v1.0 set the rebuild has re-checked S1's theorems: the Hamming equality at $n = 5$, $k = 1$, the sixteen cosets tiling the space, distance 3 and the cyclic stabiliser set **[R]**. The "v2.x canon stack" of the precedence rule is itself under revision: for the v2.0 set, Part A of this file governs. [rebuild: SEL-1, SEL-3, SEL-5]

5.2 B.2 — Pointers and quotations

Every pointer and quotation below was checked against the released v1.0 sources, the copies of the public release (an earlier copy of S2 differs in its abstract; see S2-1). Revision 1 of this file counted eleven wrong pointers in the v1.0 errata. The check finds six — those of M-3, M-4, M-5, M-7, M-8 and E7 — and one value attributed to the wrong place (S2-1); the other rows it listed were right but coarse (M-2, S1-2, S4-1) or gave no section at all (S1-1, S2-2, S5-4). The corrections that some of these rows carry wait for the rebuild (B.3); only their pointers and quotations are corrected here.

row	the v1.0 errata prints	correction
M-1	**[DRIFT]** §3, row M-1: "§7.1 "The snap: $K_5 \to K_4$+ emitted bit""	**Not a quotation.** The preprint does not print "emitted bit". §7.1 says that §7.1: "the anchor decouples from the bulk ($K_5 \to K_4 + \text{bit}$)", and App. B lists App. B: "Snap mechanism ($K_5 \to K_4 + \text{bit}$)". The row's correction (E1) waits (B.3).
M-3, E7	**[DRIFT]** §3, row M-3: "§5 8ς bond bookkeeping" §2, row E7: "(S5 abstract; preprint §5)"	**§8.1, not §5.** §5 is the section on the gauge couplings; the bookkeeping is in §8.1: §8.1: "The factor 8ς comes from 4 bonds $\times$ 2 endpoints, each contributing strain ς." The correction itself (E7) waits (B.3).
M-4	**[DRIFT]** §3, row M-4: "§6 π-in-ϖ gloss"	**§4.3, not §6**, as E8 already says: §4.3: "π is the $U(1)$ phase space factor from the gauge sector".
M-5	**[DRIFT]** §3, row M-5: "§9 τ_p heuristic"	**§8.3, not §9**, as E6 already says: the heuristic $\tau_p \sim 1/\varsigma^4 \sim 10^{38\text{–}40}$ years is printed in §8.3 (Proton Stability); §9 is the section on open problems.
M-7	**[DRIFT]** §3, row M-7: "§8.2 radion trace coupling"	**§7.1, not §8.2.** The vertex is printed once, in §7.1: §7.1: "The radion couples to matter via the stress-energy trace:"; §8.2 gives only the branching ratios.
M-8	**[SUPERSEDED]** §3, row M-8: "§3.5 dual-route strain ("K_4 and C_5 independently produce the same strain")"	**Withdrawn as an erratum of the preprint.** The preprint has no §3.5 (its §3 ends at §3.4) and prints neither a dual route nor the formula $3/8 - \text{Hb}_4$. The quoted phrase is TN-1's: §4.2: "K_4 and C_5—the two graphs in the code – independently produce the same strain." The preprint's strain is the boxed $\varsigma = 3/(80\varphi^2)$ of §4.2, labelled Constructed in its Table 2. The pointer came from TN-1 (TN1-4); what the row's demotion of the K_4 route means for the strain waits (B.3).
M-10	**[SUPERSEDED]** (a row to add) §6.5: "This is defined purely from information theory (no M_{Pl}), yet agrees with $\mu_0 = 25.34$ GeV to 0.6% – a nontrivial consistency check."	**Not a check.** A pure number can be compared with an energy only through a unit: the agreement holds in GeV and fails in any other unit, and the preprint derives no unit. The same holds for S2's "two independent routes" (S2-3). Revision 1 of this file drafted the row.

row	the v1.0 errata prints	correction
E5, M-11, S1-2, S4-1	**[DRIFT]** §2, row E5: "α^{-1} deviation quoted unsigned ("22 ppm"; S1 §9, S4 abstract and figures)"	**More prints than listed.** In the preprint: abstract: "$\alpha^{-1} = 137.033$ (22 ppm),", Table 1 (§1.1) and §5.1: "**Accuracy**: 22 ppm compared to". In S1: §9.1 and the table of §9.4. In S4: the abstract, the caption of the waterfall figure (§5; one figure, not several), the accuracy equation of §5.1 and §5.1: "The 22 ppm residual", and the summary table of §9. The sign and value of the deviation wait (B.3).
S2-1	**[DRIFT]** §5, row S2-1: "prints $\xi = 1.00651$ (abstract; boxed eq.)"	**Four values, and not in the abstract.** The released S2 prints ξ four ways: abstract: "yielding the dressing factor $\xi = 1.00636$ GeV"; 1.00651 GeV in the boxed definition and the figure of §5.2; 1.00644 GeV in §7.3; and §5.3: "The dressing factor $\xi = 1.007$ GeV", as in the tables of §5.4, §10 and §11. The 1.00651 of the abstract is in an earlier copy of S2, not in the released one. Which value is right waits (B.3).
S2-2	**[DRIFT]** §5, row S2-2: "flow figure carries a "dressed" $m_W = 81.09/\xi = 80.57$ GeV"	**Not printed so.** The flow figure (§1.1) prints $m_W = 80.57$ GeV under an arrow labelled $\times 16/5$ from μ_0. "Dressed" and "$81.09/\xi$" are the v1.0 errata's reconstruction, not the figure's. The row's verdict waits with S2-1 (B.3).
S2-3	**[DRIFT]** §5, row S2-3: "presents $\nu_0 = \text{Ħ}_4/\varsigma = 25.1802$ as the "dimensionless" radion VEV, forming a second route to the scale with ξ recording a 0.639% two-route agreement"	**Recomputed values, not prints.** S2 prints $\nu_0 = 25.178$ (§5.1) and 25.18, never 25.1802, and the agreement as 0.65%, never 0.639%: §5.2: "deviates from 1 GeV by only 0.65%". The rule of §5.4 is printed: §5.4: "Never approximate $\xi = 1$; carry the exact value." That a pure number cannot confirm an energy holds as in M-10; the rest waits (B.3).
S4-3	**[DRIFT]** §7, row S4-3: "loop coefficients under "Constructed""	**No such label in S4.** The word "Constructed" does not occur in S4. The row's **[C+]** for the loop coefficients stands as what it is, an assignment of the v1.0 errata's retrofit.
S5-1	**[DRIFT]** §8, row S5-1: "title/framing "syndrome energy observable at the LHC""	**The abstract, not the title** (the title is "Syndrome Prediction and Proton Stability"): abstract: "the syndrome energy $E_{\text{syn}} = 28.24$ GeV observable at the LHC". The correction (E3) waits (B.3).
S5-4	**[DRIFT]** §8, row S5-4: "S5's falsification line "dominant channel $p \to e^+\pi^0$" (no numbers)"	**§9.2.** The line is a row of the falsification table of §9.2: §9.2: "Dominant decay channel".
S7 header	**[SUPERSEDED]** §10: "compression entropy $n/\varsigma = 349.07$ nats."	**The wrong quantity.** S7's computational entropy per Planck 4-volume is $S_{\text{comp}} = n/\varsigma$ nats with n the spacetime dimension, abstract: "where $n = 4$ is the spacetime dimension", so $S_{\text{comp}} = 4/\varsigma$, the barrier the header itself quotes (§4.1). $349.07 = 5/\varsigma$ is the $n = 5$ row of S7's dimension table (§7.2), and "compression" is not S7's word. The header's other numbers wait (B.3).

row	the v1.0 errata prints	correction
S7-1	**[DRIFT]** §10, row S7-1: "self-label "speculative proposal" (pre-tag system)"	**Not S7's words.** S7 calls itself abstract: "a **speculative but quantitative** extension" and "Status: Speculative". As the row says, S7 predates the tags; its tag is the v1.0 errata's assignment.
TN1-4	**[DRIFT]** (a row to add) §4.2: "The dual-route strain derivation (Main Preprint [...] §3.5) becomes transparent"	**Void pointers.** TN-1's translation table (§10) also points $\varsigma = 3/8 - \text{Hb}_4$ to "Main §3.5" and $\alpha_{\text{tree}}^{-1} = 4\pi\varphi^4/\text{Hb}_4$ to "S4, Main". The preprint has no §3.5 and prints neither formula; the preprint (§5.1) and S4 (§2) print the tree form $6\varpi/\text{Hb}_4$. Revision 1 of this file found the table's pointers; the one in §4.2 is added here.

5.3 B.3 — Corrections that wait for the rebuild of the later parts

Revision 1 of this file drafted the corrections below. Each rests on results of the earlier parts that the rebuild has not yet reached, or on values of the earlier work that are not quoted until they are re-derived; so they are listed here, not asserted. Until a correction is checked, the v1.0 errata's row it concerns is read as unverified.

row	what Revision 1 drafted	waits for
E1, E4, S3-1	the snap restated as a process of two bonded nodes, and the ledger as the frozen record of the severed pair	the rebuild of the pair, the lattice and the transition
E3, S5-1	E_{syn} re-glossed as the snap energy; the visible events read as driven snaps of a pair	the rebuild of the transition
E5	the sign and value of the deviation of α^{-1}, -21.7 ppm	the re-derivation of the couplings (the node)
E7	the 8ς dressing counted as severed intra-node edges	the transition; the drift row on the snap energy's bookkeeping
S3-2	the equilibrium of the action restated as an "isoergic theorem"	the node and the transition
S3-3	the Mathieu form of S3 §3.7 corrected, and the exact spectrum of the clock sector	the node (its clock sector)
S3-4	the units of the Heisenberg sector of S3 §2.1	the node (its sector table)
S5-3	the factor $\alpha^2/4$ re-glossed as two independent hits	the transition
S2-1, S2-3	which value of ξ is right, the recomputed two-route percentage, and "eighteen" against "nineteen orders"	the re-derivation of the energy scale
S7 header	the computational entropy $S_{\text{comp}} = n/\varsigma$ at the spacetime dimension, and its numbers	the re-derivation of ς
M-8	the standing of the strain $\varsigma = 3/(80\varphi^2)$ and its tag	the node

5.4 B.4 — At the merge

- **Internal references.** The public text names no internal records: the override number of S2-3 becomes the date of the decision, and the earlier repository's file and knowledge-base names (the concordance's typewriter identifiers, the verification paragraph's validation folder) are removed.
- **Verification.** The verification paragraph is rewritten to say what was re-checked and by which checks (§5.1, row "§1").

- **Concordance.** The concordance of the v1.0 errata is rewritten to the numbering of the later parts as released at the merge; its entry for the axiom and code selection points to Part 1 read through Part A (row E2).
- **Revision history.** The merged errata lists the v1.0 errata's revision 1 (release v2.0, 29 August 2026), Revision 1 of this file (unpublished), this revision (release v2.0.1), and the merge.

6 Release plan

- **This release.** Revision 2 is released on its own, as v2.0.1 of the record (Zenodo, concept DOI 10.5281/zenodo.18832272). It adds this file and changes no released text. The next release, v2.1, is to carry Part 2.
- **No restoration of S8 v3.** Revision 1 planned to restore a repaired Supplement S8 as v3 at the next release. That repair states the triple equivalence and $k = 1$ as **[R]**; it is withdrawn. Until a rebuilt Part 1 supersedes them, Part 1 v6, S8 v2 and S9 v3 are read through Part A.
- **One sentence stays struck.** The §10 sentence of Part 1 struck by Revision 1 stays struck (P1-18).
- **Part B at the merge.** The rows of B.1 and B.2 are applied to the v1.0 errata [4]; the corrections listed in B.3 are applied as the rebuild of the later parts reaches them, and until then the v1.0 errata's rows they concern are read as unverified.
- **The merge.** This file and the v1.0 errata merge into one, as Revision 1 planned, at a release that carries rebuilt later parts, since the corrections of B.3 wait for them. The release and its date are the author's decision.

7 Concordance with Revision 1

Revision 1	Revision 2	what changed
S8-1	S8-1	kept; the reason restated
S8-2	S8-2	kept; "every downstream statement … stands" withdrawn (S8-3, S8-7)
S8-3	S8-5, P1-14	the generator set kept; "the count $n - 2k$ … unaffected" withdrawn; the fifth generator added
P1-1	P1-3, P1-4	the sector reading kept; "Constraint II is untouched" withdrawn
P1-2	P1-18	kept
S9-1	S9-8	"no content change" withdrawn
notation note	§2	kept, in position labels
restoration status	§6	the restoration of S8 v3 withdrawn
Part B, B.1–B.2	Part B, B.2	pointers re-checked: six wrong, not eleven; 25.1802 and 0.639% are recomputations, not prints; S2-1's abstract value is from an earlier copy
Part B, B.3	Part B, B.3	listed, not asserted: they wait for the rebuild of the later parts
—	Part B, B.1	new: the rows the corrections of Part A require (E2, M-2, S1-1, S6, the citation rule)

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[3] M. Hubka, "The Observer", Supplement S9 to Part I, v3, release v2.0 (29 August 2026), Zenodo, DOI 10.5281/zenodo.22161525.

[4] M. Hubka, Errata of the v1.0 release, revision 1, release v2.0 (29 August 2026), Zenodo, DOI 10.5281/zenodo.22161525.

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